

EFFECT OF HIGH-INTENSITY INTERVAL TRAINING ON CARDIOVASCULAR FUNCTION, $\dot{V}O_{2\text{MAX}}$, AND MUSCULAR FORCE

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ABSTRACT

Astorino, TA, Allen, RP, Roberson, DW, and Jurancich, M. Effect of high-intensity interval training on cardiovascular function, $\dot{V}O_{2\text{max}}$, and muscular force. *J Strength Cond Res* 26(1): 138–145, 2012—The purpose of this study was to examine the effects of short-term high-intensity interval training (HIIT) on cardiovascular function, cardiorespiratory fitness, and muscular force. Active, young (age and body fat = 25.3 ± 4.5 years and $14.3 \pm 6.4\%$) men and women ($N = 20$) of a similar age, physical activity, and maximal oxygen uptake ($\dot{V}O_{2\text{max}}$) completed 6 sessions of HIIT consisting of repeated Wingate tests over a 2- to 3-week period. Subjects completed 4 Wingate tests on days 1 and 2, 5 on days 3 and 4, and 6 on days 5 and 6. A control group of 9 men and women (age and body fat = 22.8 ± 2.8 years and $15.2 \pm 6.9\%$) completed all testing but did not perform HIIT. Changes in resting blood pressure (BP) and heart rate (HR), $\dot{V}O_{2\text{max}}$, body composition, oxygen (O_2) pulse, peak, mean, and minimum power output, fatigue index, and voluntary force production of the knee flexors and extensors were examined pretraining and posttraining. Results showed significant ($p < 0.05$) improvements in $\dot{V}O_{2\text{max}}$, O_2 pulse, and Wingate-derived power output with HIIT. The magnitude of improvement in $\dot{V}O_{2\text{max}}$ was related to baseline $\dot{V}O_{2\text{max}}$ ($r = -0.44$, $p = 0.05$) and fatigue index ($r = 0.50$, $p < 0.05$). No change ($p > 0.05$) in resting BP, HR, or force production was revealed. Data show that HIIT significantly enhanced $\dot{V}O_{2\text{max}}$ and O_2 pulse and power output in active men and women.

KEY WORDS Wingate test, anaerobic power, blood pressure, torque, fatigue index, endurance

INTRODUCTION

High-intensity interval training (HIIT) has recently been used as an alternative to traditional endurance training to alter cardiorespiratory fitness, as represented by maximal oxygen uptake ($\dot{V}O_{2\text{max}}$) and muscle metabolism. This regimen of training is characterized by 2–6 weeks of short duration (10–30 seconds), repeated efforts at near-maximal to supra-maximal work rates, interspersed with periods of recovery. It has been shown (20) to elicit adaptations in both oxygen-dependent and independent metabolism and is practical for many exercisers because of its minimal time commitment vs. aerobic exercise. In a series of studies (5,7), predominantly young men completed 6 sessions of sprint interval training over a 2-week period. Each session consisted of 4–7 Wingate tests separated by a 4-minute recovery. Results showed increased exercise performance (assessed via a time trial or cycling to exhaustion), decreased carbohydrate use, and greater fat use after a total of only 16 minutes of high-intensity exercise. However, $\dot{V}O_{2\text{max}}$ was unaffected in response to training, similar to previous data (1), although other studies reveal improved $\dot{V}O_{2\text{max}}$ in healthy women (19) and obese men (26). It is merited to further examine if short-term HIIT alters $\dot{V}O_{2\text{max}}$, because improved $\dot{V}O_{2\text{max}}$ is related to endurance performance and has been associated with reduced health risks in adults (10).

In healthy young adults, improved muscle oxidative capacity (6) and insulin action (4) have been revealed after short-term interval training. In obese men (26), attenuated blood pressure (BP) and waist circumference were also demonstrated, which in the long term may prevent the onset of type 2 diabetes and metabolic syndrome. Hypertension affects approximately 25–30% of American adults (17) and enhances the risk of stroke and heart attack because of chronic increased stress on the heart. Endurance training reduces resting BP, leading to decreased health risks on the order of 25–40% (25). In cardiac patients (24), interval training increased fitness and decreased subsequent health risks. In another study, only 6 days of Wingate-based training reduced systolic BP in obese men (26). Yet, it is unknown if low-volume HIIT is able to reduce BP in normotensive young men and women. This is important,

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because small reductions in BP elicit meaningful attenuations in health risks (25).

Muscle strength and endurance are necessary for completion of day-to-day activities and to ensure success in athletic competition and exercise training. Recent findings (21) show that persons with above-average muscular strength express reduced mortality compared with persons with relatively poor muscular fitness. However, no study to our knowledge has examined the effects of HIIT on voluntary force production. If shown to be effective, short-term HIIT may be employed as a means to increase force production in place of, or to accompany, traditional strength training, which may be practical for coaches and personal trainers to use in their athletes and clientele seeking to increase lower-body muscle force production.

Consequently, the aim of this study was to extend the previous findings (5,6,20,23,26) obtained in predominantly men by examining the effects of short-term HIIT on BP, $\dot{V}O_{2\max}$, and muscular force in men and women. An additional aim was to identify predictors of change in $\dot{V}O_{2\max}$ in response to HIIT, because changes in this parameter are equivocal in previous studies. It was hypothesized that (a) $\dot{V}O_{2\max}$ would be increased with HIIT, (b) resting BP and heart rate (HR) would be unaffected, (c) muscular strength and endurance would be significantly improved with HIIT, and (d) change in $\dot{V}O_{2\max}$ in response to training would be related to baseline $\dot{V}O_{2\max}$.

METHODS

Experimental Approach to the Problem

To prepare for each day of testing, the subjects were instructed to be euhydrated and well rested. To assess responses to training, the subjects completed baseline testing,

which consisted of measures of body composition and anaerobic power (day 1), $\dot{V}O_{2\max}$ and voluntary muscle force production (day 2), and substrate use (day 3) (data not reported). The HR and BP were recorded on all 3 days. Subsequently, the subjects in the training group completed 6 days of HIIT, with each day separated by at least 48 hours, followed by 2 days of posttesting, which occurred at least 48 hours after the last training day and no greater than 96 hours after the training ended. Control subjects completed all pretests and posttests 3 weeks apart but did not complete training. The subjects refrained from intense lower-body exercise and alcohol intake for 48 hours before each visit and did not eat in the 3 hours before each trial. The time of the day for all trials was standardized within subjects. The subjects were instructed to maintain current physical activity during their participation in the study.

Subjects

Twenty recreationally active men and women of a similar age, physical activity, and $\dot{V}O_{2\max}$ completed the training, and an additional 9 men and women served as controls. Subject characteristics are described in Table 1. Men were significantly ($p < 0.05$) taller and heavier than women and revealed a lower percent body fat and higher baseline BP. The control group revealed lower ($p < 0.05$) systolic BP than did the experimental group; otherwise, all demographic and physiological measurements were similar ($p > 0.05$) between groups. All the subjects completed regular exercise including aerobic and resistance training and various sports, although none were competitive athletes. They completed at least 4 h·wk⁻¹ of vigorous exercise and had done so for a minimum of 3 years. The subjects were excluded if they were obese, over 40 years old, did not meet our exercise criterion, or maintained more than one risk factor for heart disease.

TABLE 1. Baseline demographic data from all subjects.*†

Parameter	Training group		Control group	
	Men (<i>n</i> = 11)	Women (<i>n</i> = 9)	Men (<i>n</i> = 5)	Women (<i>n</i> = 4)
Age (y)	25.3 ± 5.5	25.2 ± 3.1	22.6 ± 3.1	23.0 ± 2.7
Height (cm)	177.7 ± 4.9	166.4 ± 8.0‡	175.9 ± 10.1	163.8 ± 3.9‡
Mass (kg)	80.3 ± 11.2	62.1 ± 6.8‡	77.2 ± 8.8	63.2 ± 7.3‡
Body fat (%)	11.3 ± 5.9	18.1 ± 5.1‡	10.2 ± 4.0	21.4 ± 4.0‡
PA (h·wk ⁻¹)	6.2 ± 1.9	5.4 ± 1.1	7.0 ± 3.3	9.1 ± 1.8
PA (y)	10.4 ± 6.0	14.8 ± 5.0	7.5 ± 6.1	7.6 ± 5.2
$\dot{V}O_{2\max}$ (ml·kg ⁻¹ ·min ⁻¹)	45.6 ± 4.0	41.1 ± 6.1	45.6 ± 7.2	39.1 ± 2.1
Systolic BP (mm Hg)	127.8 ± 9.7	112.6 ± 8.1‡	114.6 ± 2.4	106.9 ± 2.2‡
Diastolic BP (mm Hg)	80.3 ± 5.7	73.1 ± 5.1‡	76.4 ± 2.8	71.2 ± 2.6‡

*PA = physical activity; BP = blood pressure.

†Values are given as mean ± SD.

‡ $p < 0.05$ from men.

The subjects filled out a health-history questionnaire and provided written informed consent before participating in the study, and all experimental procedures were approved by the University Institutional Review Board.

Baseline Testing

On day 1, resting BP and HR (Polar Electro, Lake Success, NY, USA) were obtained after the subjects were seated in a quiet room for 5 minutes. The BP cuff was removed for 60 seconds, and BP measurement was repeated using manual sphygmomanometry (adult TruGage cuff, Omron Health Care, Vernon Hills, IL, USA). The test-retest correlation for resting HR and BP was equal to 0.90 and 0.98, respectively. These measures were repeated on days 2 and 3 of preliminary testing following identical procedures, with recorded values represented as the average BP and HR across the 3 days of measurement.

Body composition was estimated as the sum of 3 skinfolds (Lange, Beta Technology, Santa Cruz, CA, USA), as previously described (13,14). Skinfold measurements at the abdomen, thigh, and chest (men) and at the triceps, thigh, and suprailiac (women) were obtained, and circumferences at the hip and waist (11) to allow determination of waist-to-hip ratio (WHR). Then, subjects' height and body mass were measured.

Anaerobic power was determined using the Wingate test. After a 5-minute warm-up of unloaded pedaling on a Wingate ergometer (model 894e, Monark, Vansbro, Sweden), peak cadence ($\text{rev}\cdot\text{min}^{-1}$) was determined by requiring the subjects to pedal without resistance as fast as possible for approximately 4–6 seconds until peak cadence was attained. The subjects completed a 2-minute active recovery and were then instructed to reattain their peak cadence, upon which a predetermined resistance equal to 7.5% body weight was automatically applied to the flywheel. The subjects exercised for 30 seconds 'all-out,' and performed an active recovery for 5 minutes before completing a second Wingate test. The peak cadence for this trial was reduced by $20\text{ rev}\cdot\text{min}^{-1}$ to account for subject fatigue, and this value was never $<160\text{ rev}\cdot\text{min}^{-1}$ during all the subsequent trials. Peak, mean, and minimum power (watt and watt per kilogram) and fatigue index (percent) were recorded from all trials. To examine the effects of HIIT, data from the first familiarization trial were compared with that recorded from bout 1 on day 6 of training, similar to that in recent studies (5,7). Before the study, 5 men and women completed one Wingate test at the same time of the day over 3 separate days, to yield coefficients of variation for peak power, mean power, minimum power, and fatigue index equal to 4.61, 3.70, 4.95, and 2.70%, respectively, which are comparable with those reported in a recent study (15).

On day 2, resting HR and BP determinations were repeated, followed by incremental exercise on a cycle ergometer (Monark 828e) characterized by a 2-minute warm-up at 70 W followed by $28\text{ W}\cdot\text{min}^{-1}$ increments in

work rate until volitional fatigue, which was identified as a failure to maintain a cadence equal to $50\text{ rev}\cdot\text{min}^{-1}$. Attainment of $\dot{V}\text{O}_2\text{max}$ was confirmed using established criteria (2). During exercise, ventilation and gas exchange data were obtained breath by breath using a metabolic cart attached to a personal computer (ParvoMedics True One 2400, Sandy, UT, USA). Expired flow was measured using a Rudolph pneumotach screen, then integrated to obtain volume. Expired fractions for O_2 and carbon dioxide (CO_2) were measured using the Servomex paramagnetic O_2 analyzer and infrared CO_2 analyzer, respectively. Before exercise, the metabolic cart was calibrated with gases of known concentration ($16\%\text{O}_2$ and $4\%\text{CO}_2$) and to room air ($20.93\%\text{O}_2$ and $0.03\%\text{CO}_2$). Furthermore, a 3-L syringe was used to calibrate volume. The coefficient of variation for $\dot{V}\text{O}_2\text{max}$ was equal to 3.2%, comparable with that in other studies (12,16). During exercise, gas exchange data ($\dot{V}\text{O}_2$, $\dot{V}\text{CO}_2$, $\dot{V}_E$, respiratory exchange ratio) and HR were obtained every 15 seconds. Oxygen pulse at $\dot{V}\text{O}_2\text{max}$ was calculated from the following equation: $\dot{V}\text{O}_2\text{ (ml}\cdot\text{min}^{-1})/\text{HR (b}\cdot\text{min}^{-1})$.

After this trial, the subjects completed a 5-minute recovery of unloaded pedaling on the cycle ergometer and were then prepared for maximal knee extension and flexion of the dominant leg on an isokinetic dynamometer (Biodex System 3, Shirley, NY, USA). Straps were placed over the trunk to restrict movement to the exercising leg, and knee range of motion was determined for each subject, which required an additional 4–5 minutes. They completed 5 maximal repetitions of exercise at $60^\circ\cdot\text{s}^{-1}$ followed by 20 repetitions at $180^\circ\cdot\text{s}^{-1}$, with exercise beginning with the leg in the fully flexed position. Bouts were separated by 2 minutes of passive recovery, during which the subject remained in the dynamometer, yet the strap placed on the exercising leg was loosened. The subjects were provided strong verbal encouragement during exercise, yet they had no feedback regarding their performance during the protocol. Settings for the arm length and chair position were recorded and repeated during posttesting. Peak and average torque (newton per meter), power (watt), total work (Joules), and work fatigue (percent) were recorded for both knee extension and flexion across both bouts. Pilot testing revealed a coefficient of variation for peak extension torque, peak flexion torque, and extension total work equal to 5.3, 6.5, and 7.8%, respectively.

High-Intensity Interval Training

At least 48 hours after the last baseline trial at the same time of the day, the subjects completed their first day of HIIT, consisting of 4 Wingate tests as previously described. This regimen was repeated at least 48 hours later, and subsequent days of training over the next 2 weeks required completion of 5 (sessions 3 and 4) and 6 (sessions 5 and 6) Wingate tests, similar to in previous procedures (7). Participants were instructed to standardize their food intake in the 24 hours before each training day, which was confirmed via written

recalls and to refrain from intense exercise in the 48 hours before each visit. Strong verbal encouragement was provided to the subjects during exercise, and they were unaware of the remaining time in each 30-second exercise bout. Five minutes of active recovery in the form of unloaded pedaling was completed between bouts, although occasionally subjects were provided additional time if they were physically unable to initiate the subsequent exercise bout.

Posttesting

The subjects were instructed to maintain their current physical activity regimen during training, which was recorded via written journals. At 48 hours after the last day of training, body composition, HR, and BP were measured following identical techniques as in baseline testing. Subsequently, the subjects repeated tests of $\dot{V}O_2\text{max}$ and voluntary muscle force production following identical procedures as in baseline testing. Controls were retested 3 weeks after the completion of baseline testing.

Statistical Analyses

Data were expressed as mean $\pm$ SD and analyzed using SPSS Version 16.0 (Chicago, IL, USA). Two-way analysis of variance with repeated measures was used to examine the

differences in power output, $\dot{V}O_2\text{max}$, BP, HR, body composition, and voluntary force production, with training as a within-subjects factor and group (experimental and control) as a between-subjects factor. Data were not separated by gender, because recent data (3) reveal that responses to short-term HIIT are similar between men and women. No physiological variable with the exception of BP differed between experimental and placebo groups at baseline. A significant training $\times$ group interaction was used to identify training-induced changes in these variables. The Greenhouse-Geisser correction was used to account for the sphericity assumption of unequal variances across groups. Tukey's post hoc test was used to detect differences between means when a significant F ratio was obtained. Independent t -test was used to examine differences in demographic characteristics between men and women and the experimental and control group. Multiple regression was used to determine pairwise correlations between variables and to identify predictors of change in $\dot{V}O_2\text{max}$ in response to HIIT. Statistical significance was established as $p \leq 0.05$.

RESULTS

All participants completed all the requirements of the protocol. Current physical activity was reduced ($p < 0.05$)

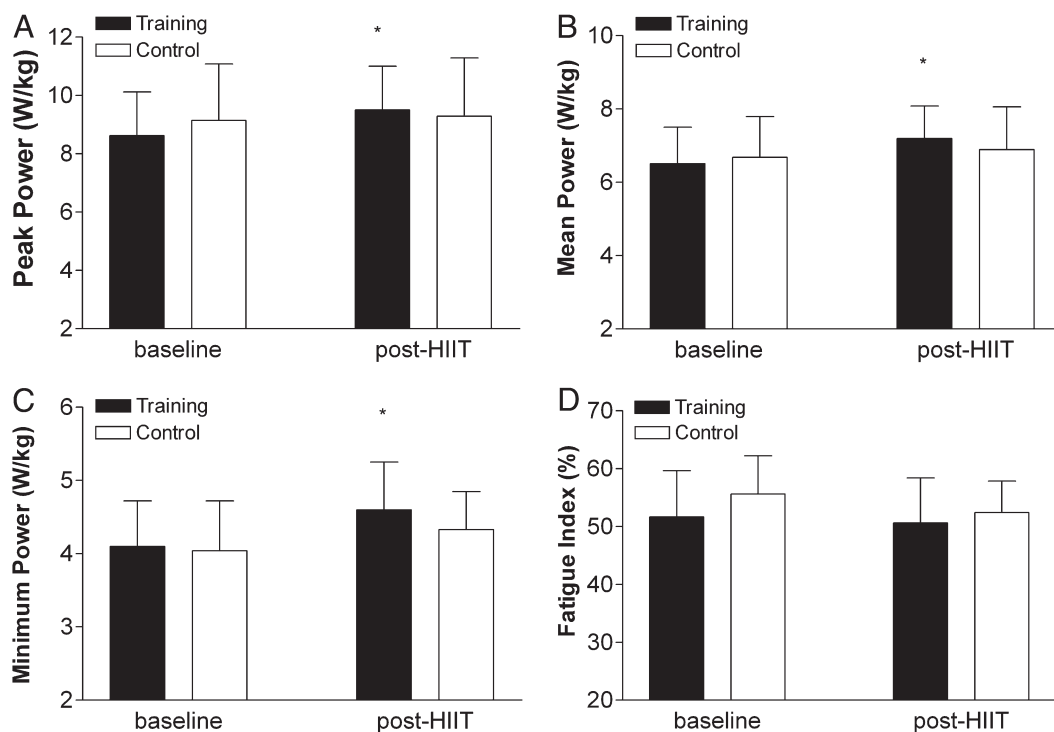


Figure 1. Change in (A) peak power, (B) mean power, (C) minimum power, and (D) fatigue index in active men and women who performed HIIT ($n = 20$) and controls ($n = 9$); *Significant interaction ($p < 0.05$) in response to HIIT.

TABLE 2. Maximal gas exchange and heart rate data for subjects who completed HIIT ($n = 20$) and controls ($n = 9$).^{*†}

Parameter	Training group		Control group	
	Baseline	Post-HIIT	Baseline	Posttest
$\dot{V}O_{2\max}$ (L·min ⁻¹)	3.1 ± 0.7	3.3 ± 0.8 [‡]	3.1 ± 0.8	3.1 ± 0.7
$\dot{V}O_{2\max}$ (ml·kg ⁻¹ ·min ⁻¹)	43.6 ± 5.4	46.0 ± 5.2 [‡]	42.7 ± 6.3	43.4 ± 5.7
$\dot{V}CO_{2\max}$ (L·min ⁻¹)	3.7 ± 0.8	3.9 ± 0.9 [‡]	3.7 ± 1.0	3.6 ± 0.9
RER _{max}	1.18 ± 0.07	1.17 ± 0.06	1.21 ± 0.09	1.17 ± 0.08
HR _{max} (b·min ⁻¹)	186.4 ± 10.0	187.2 ± 9.3	181.0 ± 13.2	181.0 ± 9.6
$V_{E\max}$ (L·min ⁻¹)	125.1 ± 32.5	128.9 ± 32.1	118.3 ± 38.2	118.1 ± 39.1
$O_{2\text{pulse}}$ (ml·beat ⁻¹)	16.9 ± 4.3	17.9 ± 4.5 [‡]	17.0 ± 4.6	17.1 ± 3.8

*HIIT = high-intensity interval training; RER = respiratory exchange ratio.

†Values are given as mean ± SD.

‡Significant interaction ($p < 0.05$).

during training (3.9 ± 1.9 h·wk⁻¹) compared with baseline (5.8 ± 1.6 h·wk⁻¹). A similar result was revealed in the control group, because current physical activity was significantly lower ($p < 0.05$) at posttesting (5.2 ± 3.3 h·wk⁻¹) than at baseline (7.9 ± 2.8 h·wk⁻¹).

Wingate-Derived Measures

Significant training × group interactions revealing improved peak, mean, and minimum power were revealed in subjects

who performed HIIT (Figures 1A–C). Mean fatigue index was unaffected in response to training (Figure 1D). Measures of power output and fatigue index were unchanged in the control group compared with baseline values.

Resting Heart Rate and Blood Pressure

There was no main effect or interaction effect for HR and BP. The resting HR was similar ($p > 0.05$) at baseline (69.0 ± 7.3 b·min⁻¹) vs. post-HIIT (67.9 ± 8.3 b·min⁻¹). Similarly,

TABLE 3. Muscle function data for subjects who completed HIIT ($n = 20$) and controls ($n = 9$).^{*†}

Parameter	Training group		Control group	
	Baseline	Post-HIIT	Baseline	Posttest
Velocity = 60°·s ⁻¹				
Peak KE torque (N·m ⁻¹)	203.6 ± 65.4	201.0 ± 63.1	186.3 ± 53.4	192.7 ± 54.4
Avg KE torque (N·m ⁻¹)	187.7 ± 61.2	187.8 ± 64.4	173.1 ± 51.3	175.8 ± 56.2
Avg KE power (W)	128.2 ± 46.8	130.8 ± 44.5	122.8 ± 46.1	127.9 ± 48.6
Peak KF torque (N·m ⁻¹)	131.6 ± 37.4	140.5 ± 43.5	124.1 ± 43.4	128.5 ± 43.4
Avg KF torque (N·m ⁻¹)	123.7 ± 37.4	132.2 ± 43.8	116.6 ± 40.1	115.2 ± 38.2
Avg KF power (W)	92.4 ± 28.5	97.5 ± 31.1	95.4 ± 37.5	98.1 ± 37.0
Velocity = 180°·s ⁻¹				
Peak KE torque (N·m ⁻¹)	128.2 ± 37.3	130.6 ± 32.8	123.9 ± 38.5	130.8 ± 41.2
Avg KE torque (N·m ⁻¹)	101.0 ± 28.2	105.7 ± 27.2	101.7 ± 31.7	106.8 ± 32.4
Avg KE power (W)	175.1 ± 54.6	187.7 ± 52.5	178.9 ± 63.9	187.3 ± 67.0
KE work fatigue (%)	42.5 ± 8.9	39.2 ± 6.0	42.8 ± 8.2	39.9 ± 7.7
Peak KF torque (N·m ⁻¹)	101.2 ± 30.6	107.6 ± 30.6	96.6 ± 30.5	100.4 ± 32.0
Avg KF torque (N·m ⁻¹)	83.5 ± 25.4	87.9 ± 25.8	79.8 ± 24.2	86.8 ± 28.2
Avg KF power (W)	142.9 ± 49.2	151.1 ± 47.3	139.4 ± 47.5	148.1 ± 54.5
KF work fatigue (%)	38.0 ± 8.1	37.9 ± 7.4	38.7 ± 6.3	39.3 ± 5.8
KE total work (J)	1,584.5 ± 520.7	1,677.6 ± 507.1	1,562.2 ± 560.3	1,658.9 ± 555.5
KF total work (J)	1,357.0 ± 489.3	1,416.7 ± 479.7	1,314.0 ± 460.5	1,362.6 ± 451.0

*KE = knee extension; KF = knee flexion; Avg = average.

†Values are given as mean ± SD

there was no change in systolic (120.9 ± 11.7 vs. 121.2 ± 8.9 mm Hg) or diastolic BP (77.1 ± 6.5 vs. 76.5 ± 6.4 mm Hg) in response to HIIT. Compared with baseline, the control group revealed no change in HR (64.2 ± 9.8 vs. 63.9 ± 9.9 b·min⁻¹), systolic BP (111.2 ± 4.6 vs. 111.0 ± 8.5 mm Hg) or diastolic BP (74.1 ± 3.8 vs. 71.3 ± 7.7 mm Hg).

Gas Exchange Data

Maximal gas exchange data for all subjects are revealed in Table 2. A significant training $\times$ group interaction demonstrated increases in $\dot{V}O_{2\max}$, $\dot{V}CO_{2\max}$, and maximal O_2 pulse in the training group. The magnitude of change in $\dot{V}O_{2\max}$ ranged from 0 to 20% (mean = $6.3 \pm 5.4\%$). Control subjects revealed no change ($p > 0.05$) in $\dot{V}O_{2\max}$ or O_2 pulse at posttesting compared with baseline values.

Muscle Force Production

Compared with baseline, peak and average extension torque and average power at $60^\circ \cdot s^{-1}$ were unaltered ($p > 0.05$) with training (Table 3). Although significant changes in additional parameters of knee extensor and flexor muscle function were demonstrated from baseline to posttesting (main effect = $p < 0.05$), no significant training $\times$ group interaction was revealed for any variable.

Body Composition Data

Percent body fat (14.3 ± 6.4 vs. $14.2 \pm 6.4\%$), WHR (0.77 ± 0.06 vs. 0.76 ± 0.07), and body mass (72.1 ± 13.1 vs. 72.3 ± 13.2 kg) did not change with HIIT. These variables were also unaltered ($p > 0.05$) in control subjects (data not reported).

Correlation Analyses

Significant pairwise correlations were revealed between change in $\dot{V}O_{2\max}$ and baseline $\dot{V}O_{2\max}$ ($r = -0.44$, $p = 0.05$) and baseline fatigue index ($r = 0.50$, $p < 0.05$), yet there were no relationships ($p > 0.05$) between change in $\dot{V}O_{2\max}$ and baseline power output ($r = -0.08$ to -0.33) from the Wingate test. When 2-predictor models were developed to explain change in $\dot{V}O_{2\max}$ in response to HIIT, the combination of fatigue index and baseline $\dot{V}O_{2\max}$ revealed a significant model ($R = 0.62$, $R^2 = 0.39$, $SEE = 4.50$, $p < 0.05$), with both variables serving as significant independent predictors of change in $\dot{V}O_{2\max}$.

DISCUSSION

The primary aim of this study was to examine the effects of short-term HIIT on BP, cardiorespiratory fitness, and muscular force in active men and women, and identify predictors of change in $\dot{V}O_{2\max}$. Results revealed that training did not alter resting BP or HR, although $\dot{V}O_{2\max}$, $\dot{V}CO_{2\max}$, and O_2 pulse were significantly enhanced with HIIT. The magnitude of increase in $\dot{V}O_{2\max}$ was related to baseline fatigue index and $\dot{V}O_{2\max}$ but not peak or mean power output. Peak, mean, and minimum power output were significantly higher with training, although various measures of voluntary strength and endurance were unaltered. Data

reveal that short-term HIIT improves $\dot{V}O_{2\max}$, power output, and O_2 pulse in active men and women, although it is unknown if sustained HIIT would continue to induce these adaptations in the long term.

The increase in $\dot{V}O_{2\max}$ observed in this study is opposed by that of previous studies that employed a similar regimen of short-term HIIT in young men and women. Burgomaster et al. (5,7) revealed no change in $\dot{V}O_{2\max}$, similar to early data (23) in men completing sprint training for 8 weeks. Baseline $\dot{V}O_{2\max}$ of these subjects was greater (48.0 – 49.0 ml·kg⁻¹·min⁻¹) than that of our subjects. In this study, 3 active women ($\dot{V}O_{2\max} = 45.2$ – 49.9 ml·kg⁻¹·min⁻¹, fatigue index = 35–42%) also revealed no change in $\dot{V}O_{2\max}$, although they demonstrated improved power output, muscle endurance, and higher fatigue index with HIIT. It is possible that there is a maximum $\dot{V}O_{2\max}$ value at which short-term interval training does not improve $\dot{V}O_{2\max}$ in active, young men and women. In sedentary men and women completing 3–6 weeks of sprint interval training (6), $\dot{V}O_{2\max}$ was improved by 7%, similar to that of this study but less than that reported in active women (+13%) completing 7 sessions of HIIT (22) and young men (11%) completing 14 sessions of repeated 15- to 30-second sprints (20). When our subjects were separated by fatigue index ($>50\%$, $n = 12$; $<50\%$, $n = 8$), change in $\dot{V}O_{2\max}$ was greater ($8.9 \pm 5.5\%$) in subjects with higher fatigue index vs. those with a low fatigue index ($2.4 \pm 2.1\%$). In addition, 37% of the change in $\dot{V}O_{2\max}$ was explained by baseline $\dot{V}O_{2\max}$ and fatigue index. Esbjornsson-Liljedahl et al. (9) reported that women, and not men, displayed increased type IIb fiber size in response to repeated Wingate tests. Persons with a greater percentage of type II fibers typically express a higher fatigue index, because type II fibers contain few mitochondria. In contrast, individuals with more type I fibers may reveal improved power output and muscle force production, and not improved $\dot{V}O_{2\max}$, in response to short-term interval training.

The mechanism explaining this increase in $\dot{V}O_{2\max}$ is beyond the scope of this study but may be because of enhanced cardiac function or O_2 pulse, as was observed posttraining. Recently, Daussin et al. (8) reported increased maximal stroke volume and cardiac output in response to 8 weeks of interval training, but not aerobic training, in sedentary men and women. Further investigation is needed to confirm these findings in other populations.

Resting BP and HR obtained over 3 days of baseline testing and 2 days of posttesting were unaltered in response to high-intensity training. These findings are similar to data from Rakobowchuk et al. (18) in which untrained subjects completed 6 weeks of interval training. In obese men with elevated resting BP, significant reductions in systolic BP were evident in response to 2 weeks of Wingate-based training (26). These data suggest that normotensive subjects may not reveal reduced resting BP in response to short-term HIIT, yet it may be effective in individuals with high BP.

Enhanced mean or peak power output is commonly observed in the majority of studies (5–7,26) employing short-term interval training in young men and women. Our magnitudes of increase in peak power (10.1%), mean power (10.6%), and minimum power (12.4%) are comparable with previous investigations. However, Rodas et al. (20) revealed no change in power output in response to 14 sessions of interval training. It is likely that discrepancies in the specific identification of peak power, subject characteristics, cycle ergometer used, and duration of training explain these differences across studies.

To our knowledge, no study has examined changes in voluntary muscle force production in response to short-term HIIT, although 1 study (23) showed improved isometric strength in men performing repeated treadmill sprinting for 8 weeks. Results demonstrated minimal changes in force production or fatigability of the knee extensors and flexors. This result was surprising, based on the documented improvements in mitochondrial function (6) that occur with sprint interval training. It is plausible that completion of additional sets or repetitions of isokinetic dynamometry may have been needed to detect changes in force production in response to HIIT. Moreover, subjects were only allotted 10 minutes of recovery between $\dot{V}O_2\text{max}$ assessment and the isokinetic protocol, and it is possible they completed knee extension and flexion exercise while they were still fatigued.

There are a few limitations to this study. Data can only be generalized to active men and women free of known diseases. Although improved $\dot{V}O_2\text{max}$ and power output were observed with training, it is unknown if these can be sustained when training ensues for >2–3 weeks and if the familiarization trials completed before training contributed to these adaptations. Additionally, no mechanistic explanations can be identified for the changes in fitness observed in this study, although enhanced mitochondrial enzyme activities (citrate synthase, B-hydroxy acyl dehydrogenase, and pyruvate dehydrogenase) (6) and muscle adaptations (reduced phosphocreatine [PCr] degradation, enhanced glycogen content) (6) and increased type IIa ratio and reduced IIb ratio (1) have been repeatedly demonstrated in response to low-volume sprint interval training. The control group was smaller than the experimental group, although demographic and physiological characteristics between groups were similar with the exception of resting BP. However, this study is strengthened by inclusion of men and women similar in age, cardiorespiratory fitness, and physical activity.

In conclusion, data suggest that 2 weeks of HIIT improves $\dot{V}O_2\text{max}$ and peak and mean power output, yet it has no effect upon resting HR, BP, or muscle force production. The magnitude of change in $\dot{V}O_2\text{max}$ and peak power was significantly related to baseline $\dot{V}O_2\text{max}$ and fatigue index, suggesting that adaptation to interval training may be related to subjects' baseline cardiorespiratory fitness and fatigability.

PRACTICAL APPLICATIONS

It is recommended that persons exercise 30–60 min·d⁻¹ to reduce health risks, improve fitness, and promote weight loss, yet time is often a barrier to exercise adherence. These data reveal that only 6 days of interval-based training improves lower-body power output and aerobic fitness in young, recreationally active men and women. However, neither muscular strength and endurance of the quadriceps or hamstrings nor HR and BP were modified in response to interval training. Individuals desiring to improve aerobic fitness and leg power might incorporate short-term interval training into their current exercise regimen, because it is a powerful stimulus to promote various adaptations that may improve performance and overall health status.

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